

Hyperledger Sawtooth

Sawtooth is an open source blockchain platform founded under Hyperledger. It facilitates a modular and flexible platform for implementing transactions based updates to a shared state between untrusted parties coordinated by consensus algorithms. It simplifies blockchain application development by separating the core system from the application domain. It allows flexibility to the applications to choose the transaction rules, permissioning, and consensus algorithms that support unique business needs. Sawtooth is built to solve the challenges of permissioned (private) networks. Clusters of Sawtooth nodes can be easily deployed with separate permissioning. There is no centralised service that could potentially leak transaction patterns or other confidential information.

Hyperledger Sawtooth supports creating and broadcasting events. This allows applications to:

- Subscribe to events that occur related to the blockchain, such as a new block being committed or switching to a new fork.
- Subscribe to application-specific events defined by a transaction family.
- Relay information about the execution of a transaction back to clients without storing that data in state.

Figure 4 highlights the Hyperledger Sawtooth architecture.

Hyperledger Indy

Hyperledger Indy is another distributed ledger that has been especially designed for decentralised digital identity management. It is termed a public permissioned blockchain network for identity, with privacy features. With Indy, identity owners take full control of their personal data in order to expose only selected information or avoid exposing personal data. It provides advanced security features in terms of privacy, data breaches and identity theft.

It is composed of the following projects.

- **Indy Node:** This is an Indy-Plenum based implementation of the distributed ledger with Redundant Byzantine Fault Tolerance (RBFT).
- **Indy-sdk:** This is a client library to interact with the Indy network.
- **Indy-crypto:** This is based on the Apache Milagro Cryptographic Library (AMCL) and supports C, RUST and the Python API.

Figure 5 highlights the Hyperledger Indy architecture.

Hyperledger tools

Hyperledger Caliper

Hyperledger Caliper is a benchmarking tool for blockchain as it allows users to measure the performance of blockchain implementation with predefined use cases. It provides numerous reports, and contains performance indicators when used with varied blockchain solutions like Hyperledger Besu, Burrow, Ethereum, Hyperledger Fabric, Iroha and Sawtooth.

The benchmarking parameters in Caliper are as follows.

- **Success rate:** It allows end users to determine how many out of all submitted transactions have been processed successfully and written on the blockchain.

- **Throughput:** This reports the number of transactions per second processed by the underlying blockchain network.
- **Latency:** This measures the rate at which the transactions are processed and written on the ledger.
- **Resource consumption:** This report highlights how many resources are consumed by each Docker container.

Hyperledger Ursa

Ursa is a shared cryptographic library that enables people to avoid duplicating other cryptographic works and enhance overall security. Ursa provides open source blockchain developers with reliable, secure, and pluggable cryptographic implementations.

The major artifacts of Ursa are:

- C-callable library interface
 - Rust Crate
- Ursa is divided into two sub-libraries: Libursa and Libzmix.
- **Libursa:** This has been designed for cryptographic primitives like digital signatures, encryption and key exchange.
 - **Libzmix:** This is a significant way to create zero-knowledge proofs, proving statements about multiple cryptographic building blocks that contain encryption and the signature. 

References

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- [4] <https://sawtooth.hyperledger.org/>
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- [6] <https://www.hyperledger.org/use/caliper>
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